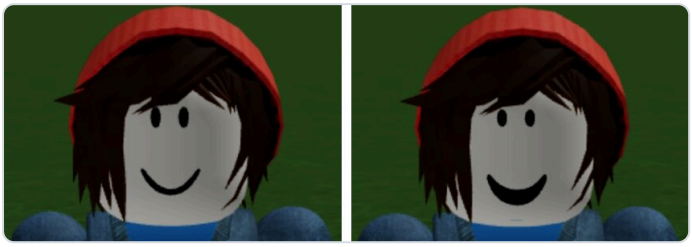
ROBLOX STUDIO EXPERT SKILLS · TEAM SKILL 10 OF 14

Animation and Characters

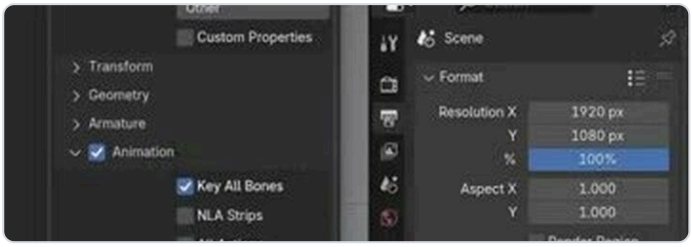
Technical animator, character TD

`scenario-roblox-animation`

Builds, layers and plays Roblox character, NPC and creature animation in code, then proves each clip, IK chain and face by measured joints on server and client.



Live playtest, Studio 0.743: neutral head, then a face clip holding JawDrop 0.55. 23 of 23 tests pass.



Blender FBX export: armature only, Key All Bones, NLA Strips and All Actions off, 24 fps. Mykaiju, 00:25:51.

USE IT WHEN

- Animation not playing, the wrong clip wins, or legs freeze while walking
- Importing a Blender or FBX clip, a custom rig, or a creature
- Hit timing from markers, IK look-at or foot planting, dynamic head faces

WHAT THE EXPERTS INSIST ON

- Create IKControls on the server and set look-at Weight to 0.999: at 1 the neck twists when the target passes behind.
  Suphi Kaner (scripting educator)
- Time gameplay with animation markers, never waits: one event name, with the detail passed as a parameter. Develuper (developer educator)
- Export Blender clips at 24 fps, armature only, no scale keys: NLA strips and All Actions off, Force Start/End Keying on.
  Mykaiju (technical animator)
- Animating at 60 fps doubles the keys for no gain in gameplay; 24 fps is enough. Nisky (Blender animation educator)
- Anchor NPC rigs and disable their Animators out of range: NPC animation is a real server cost.
  Peter McNeill (Roblox performance engineer)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Inspect the rig. Gate:** Joint classes known; every animated part has a path from the root; Animator exists.
2. **Build or import clips. Gate:** Stepped joints match keys within 0.3 degrees; Blender tails within 0.01 studs.
3. **Layer and time. Gate:** The intended track wins each joint; each marker fires once per cycle.
4. **Preview in Edit. Gate:** Pose sheet read, frames differ, no posed joint left before saving.
5. **Playtest both sides. Gate:** Server finds the client's track; both sides report the same joint angle.
6. **Layers, IK and faces. Gate:** Neck yaw, IK gap and face values measured on the server, each captured.
7. **Audit and publish list. Gate:** AnimAudit clean; no temporary animation id left in shipped code.

KEY NUMBERS

Marker at 0.25 s of playback	**fired at 266 ms, frame 16.3 ms**
Fade from Play(0.4)	**weight rises 0.25 per 0.1 s**
IK reach, reachable target	**hand 0.20 to 0.23 studs away within 0.1 s**
Face write or tween, player head	**back to 0 within 0.5 s; a face clip holds**
Blender to Studio, 10 bones, 25 frames	**worst bone-tail error 0.0002 studs**
Raw-import clip after wrapping the rig	**still binds: jaw moved 40 degrees (live)**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- AnimAudit.all returns no error and no warning; no temporary id ships.
- Every played track reports Length above 0, intended priority, WeightCurrent 1 at rest.
- Server and client report the same joint angles for replicated tracks.
- Pose sheet of each clip at 4 to 6 times, judged: silhouettes, arcs, closed loop.

HANDOFFS

Receives Imported rigs, skinned meshes and FBX or Blender motions from scenario-roblox-assets, including rigs and motions from scenario-3d.

Delivers Clips with markers at contact frames to gameplay; marker names and times to vfx-audio; a publish list for the user.

TOOLKIT `RBX_ANIMATION.PY`

- `pose_sheet()`: steps a track in Edit, captures, one contact sheet
- `blender()`: headless Blender audit, FBX export, re-import, samples
- `run()`: Studio chunk with AnimRig, AnimClip, AnimIK, AnimFace kits

MAIN EXPERT SOURCES

Suphi Kaner
scripting educator (unverified), Inverse Kinematics, HingeConstraint: Roblox Scripting Tutorial

Develuper
developer educator (unverified), Make Your Animations 10x Better with Animation Events

Mykaiju
technical animator (unverified), How to animate and import animations from Blender to Roblox

Nisky
Blender animation educator (unverified), Roblox Animation in Blender: Advanced Guide (2026)

STATUS

Tests offline O1 to O5 pass

In app 23 of 23 live tests pass in Roblox Studio 0.743

G7 animation and characters

without skills	8/15
with skills	14/15